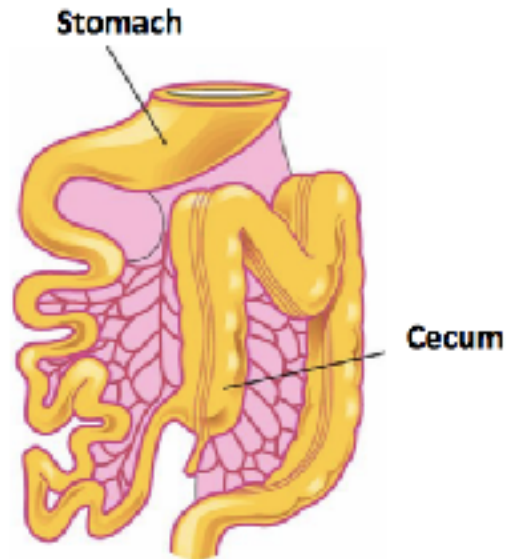
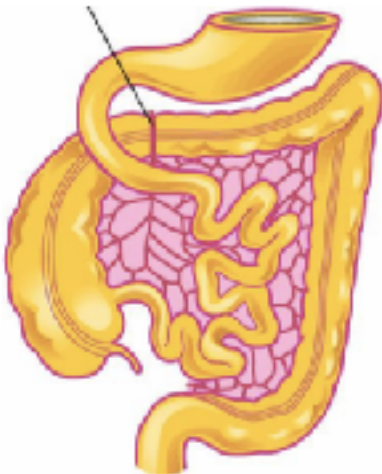


1. Midgut develops outside of the abdomen, then rotates back in
*Normal: counterclockwise. Cranial limb returns first, to center left, cecum/appendix to bottom right. Possible anomalies...

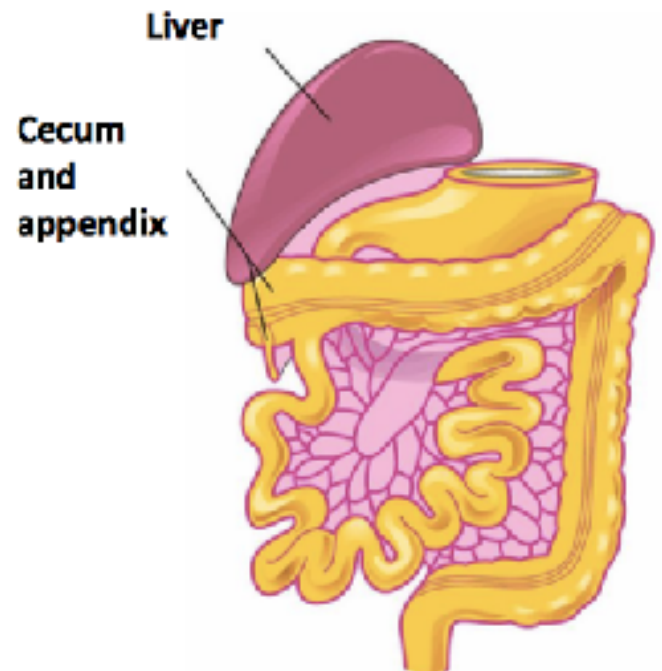
- a. Nonrotation: Caudal limb goes in first. Cecum ends up in between the small and large intestine, instead of to the far right

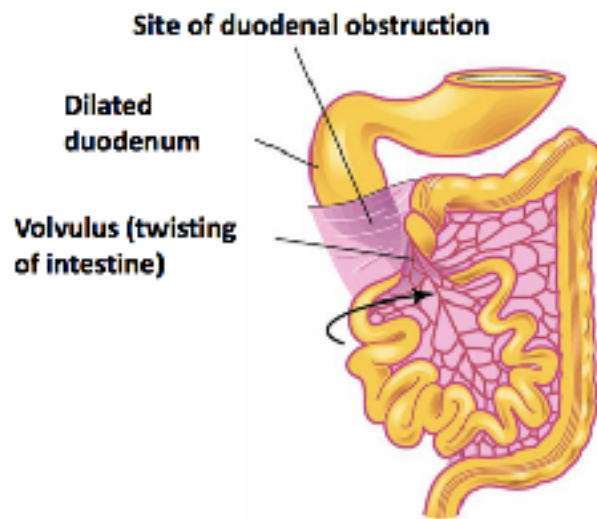
**Superior mesenteric artery
(compressing transverse colon)**



- b. Clockwise rotation. Duodenum ends up in front of transverse colon instead of behind

- a. Subhepatic cecum/appendix. Cecum gets stuck to liver and doesn't descend to iliac fossa. Can make it hard to diagnose appendix because pain is in the wrong place

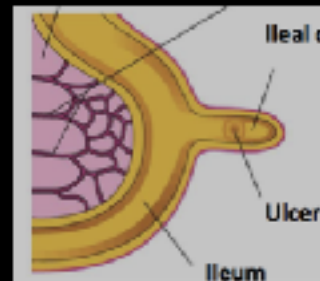




d. Mixed rotation and Volvulus:
Twisted up intestine (last 90° of
rotation fucked up)

Meckel's Diverticulum

- Remnant of the omphaloenteric duct projecting off the ileum
- Rule of 2s
 - 2ft proximal to ileocecal jxn
 - 2% of population
 - 2x as likely in males
 - 2 inches long
 - 2 tissue types (pancreatic and gastric)
 - Mimics 2 conditions (Appendicitis and Peptic Ulcer)



Umbilical Hernia

Covered by: Skin

Incomplete closure of umbilical ring



Omphalocele

Covered by: Fetal membranes

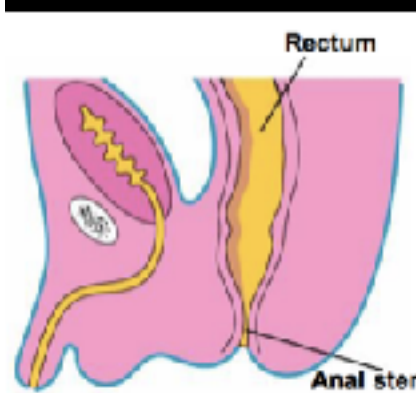
Persistence of Abdominal Herniation



Gastroschisis

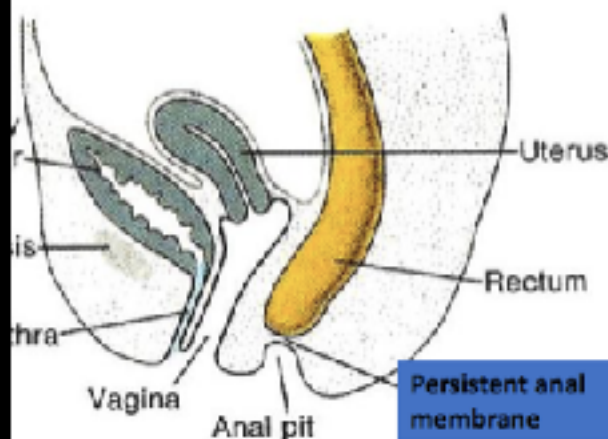
Covered by: Nothing

Incomplete closure of lateral folds (wk 4)



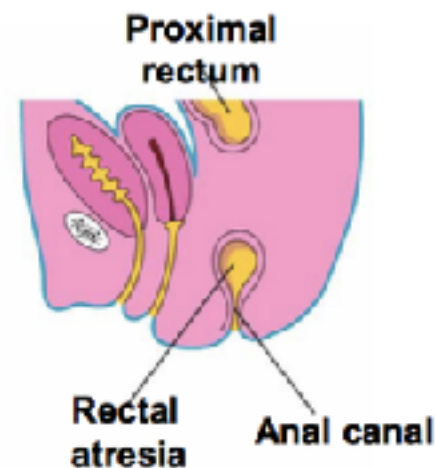
Anal Stenosis

Dorsal deviation of Urorectal Septum



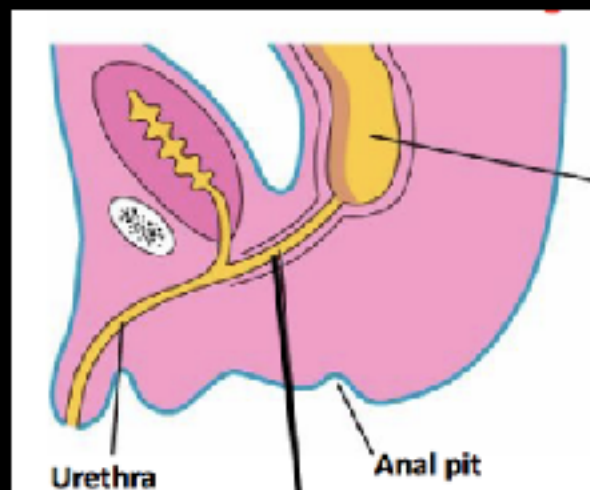
Membranous Atresia

Failure of Epithelial Plug to perforate (week 8)



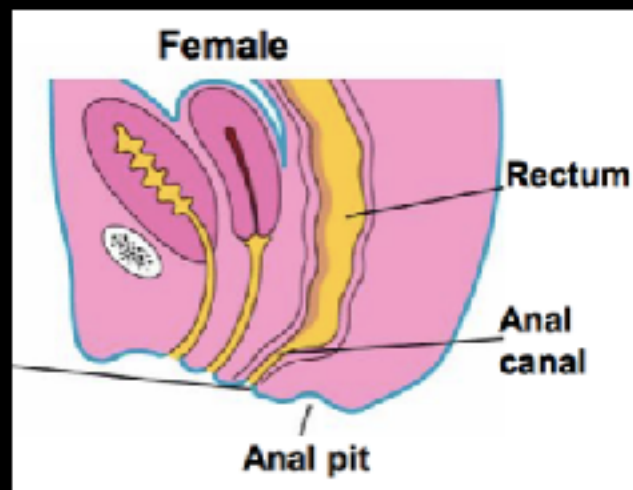
Rectal Atresia

Abnormal recanalization/loss of blood supply



Rectal Aggenesis with Fistula

Incomplete separation of the cloaca from the urogenital sinus by the urorectal septum



Imperforate Anus

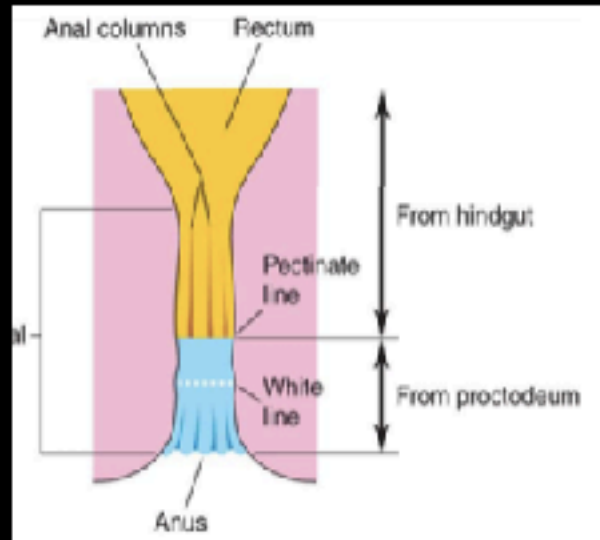
Anus: Pectinate Line

- Above Pectinate Line (upper 2/3)

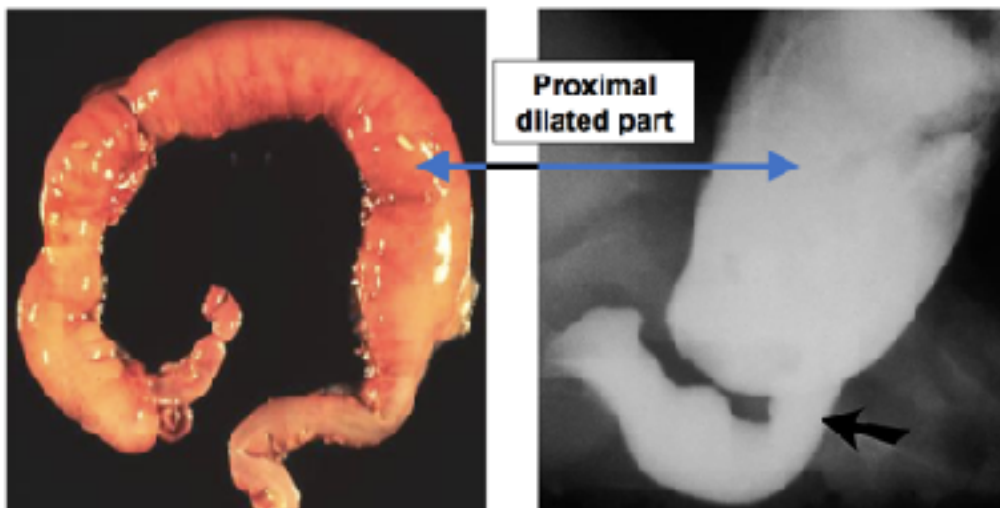
- From hindgut
- Blood supply: IMA
- Innervation: Autonomic

- Below Pectinate Line (lower 1/3)

- From proctodeum
- Blood supply: Inferior Rectal Artery (from Internal Pudendal)
- Innervation: Inferior rectal nerve (from Pudendal)



Hirschsprung Disease = Congenital Megacolon



*Neural crest cells don't migrate to chunk of large intestine. No myenteric plexus —> no peristalsis —> food gets stuck. Constipation/ distended abdomen

